INDIVIDUAL CONTRIBUTOR DOSSIER • TEAM LEAD & HEAD OF RESEARCH & PRODUCT STRATEGY

Khushi Patil

Team: Reconstruct.ai | **Institution:** JSPM RSCOE, Pune

Event: We Code Coders Hackathon

Domain: Product Strategy, Cognitive Problem Taxonomy & Milestone Management

Email: khushi@reconstruct.ai

Repository: github.com/Dhruvplaz-web/reconstruct-ai

1. Personal Mandate & Individual Contribution Statement (Rule 14)

As the Team Lead and Head of Research, I am responsible for overall project orchestration, sprint prioritization, problem validation, and ensuring 100% adherence to the hackathon's 11 mandatory submission items and 100-point judging rubric. I spearheaded the qualitative audit of developer turnover dynamics across Reddit r/ExperiencedDevs and Hacker News, formulated the theoretical foundation around Nonaka's SECI model (tacit vs. explicit knowledge breakdown), and established the multi-domain problem taxonomy spanning Enterprise Tech, Academic Research Labs, Student Clubs, and Open-Source Communities.

2. Subsystem Ownership & Technical Architecture

- **SECI & Agile Paradox Research Framework:** Theoretical formulation explaining why agile sprint speed directly accelerates organizational forgetting.
- **Competitor Teardown Matrix:** Competitive benchmarking against Glean, Atlassian Rovo, Swimm, and NotebookLM to isolate market gaps.
- **Sprint Planning & Milestone Governance:** Tracking task breakdown across all 4 team members for Rounds 1, 2, and 3.
- **User Acceptance Criteria & Success Rubric:** Defining measurable KPIs (80% ramp-up reduction, 15x leaver documentation friction cut).



Figure 1: End-to-End Operational Lifecycle Managed Under Khushi Patil's Product Roadmap

3. Technical Deliverables Across All 3 Rounds

Hackathon Round	Personal Responsibilities & Concrete Engineering Deliverables
Round 1: The Plan (Current Phase)	Authored problem understanding dossier, SECI taxonomy, competitor autopsy matrix, team responsibility breakdown, and ensured full compliance with all 11 hackathon brief requirements.
Round 2: The Build (Working Functional MVP)	Lead User Acceptance Testing (UAT), coordinate between frontend and backend leads, curate real-world test scenarios, and ensure working MVP runs end-to-end for judge evaluations.
Round 3: The Finish (Production Polish & UAT)	Direct the final 3-minute product walkthrough video, author the comprehensive security and accessibility audit documentation, and manage the delivery of the 3 major promised Round 3 product improvements.

4. Competitor Failure Analysis & Defense for this Domain

Traditional tools fail because they treat handover as a documentation task rather than an organizational sociology challenge. Confluence fails due to the 'Graveyard Effect' (wikis rot in 30 days). Glean fails due to the 'Passive Search Gap' (new hires don't know what to ask). Reconstruct.ai overcomes these structural market failures by shifting from manual push-based wikis to automated pull-based graph synthesis.

5. Judge Viva & Technical Decision Defense (Rule 16 Preparation)

Per Rule 16 ('Judges may ask any team member to explain the product, demonstrate a feature, justify a technical decision or verify their contribution'), this section details the anticipated questions and defensible justifications:

Anticipated Judge Question / Challenge	Defensible Technical Justification & Verification Evidence
Q1: Why focus on student clubs and research labs in addition to enterprise software teams?	Because institutional knowledge loss is actually more severe in non-corporate environments: student clubs face a 100% leadership reset every 1-4 years, and academic labs suffer from post-doc rotation reproducibility crises. Reconstruct.ai's architecture is domain-agnostic.
Q2: How did you ensure the team's plan is feasible within the hackathon time constraints?	We explicitly separated our Round 2 Scoped MVP (GitHub + Slack ingestion, DAG extraction, and 60-second review) from Round 3 enhancements (advanced NLI contradiction radar and ABAC toggle). We also added the 1-Click Synthetic Playground so judges can evaluate the app in 30 seconds.
Q3: How do you measure whether Reconstruct.ai actually solves the knowledge handover problem?	Through three quantitative KPIs: 1) Successor time-to-first-commit reduced from 4 weeks to under 4 days; 2) Bus Factor = 1 orphan code eliminated; 3) Leaver offboarding burden cut from 20 hours to under 60 seconds.

6. Full Project Team Roster & Peer Verification

Member Full Name	Assigned Role	Institution	Project Verification Status
Khushi Patil	Team Lead & Head of Research	JSPM RSCOE, Pune	**VERIFIED (Team Lead)**
Dhruv Patil	AI Development & Core Engineering Lead	JSPM RSCOE, Pune	**VERIFIED (AI Core Lead)**
Yuvraj Kadam	Lead Full-Stack Frontend & UI/UX	JSPM RSCOE, Pune	**VERIFIED (Frontend Lead)**
Nimisha Joshi	Systems Backend, Ingestion & Security	JSPM RSCOE, Pune	**VERIFIED (Backend Lead)**

Primary Project Repository: https://github.com/Dhruvplaz-web/reconstruct-ai • **Target Event:** We Code Coders Three-Round Hackathon